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Joint optimization of altitude and resource allocation for aerial cooperative short packet communication system

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ABSTRACT

Unmanned aerial vehicle (UAV) is considered as an important component of 6G Internet of Things (IoT) because of its deployment flexibility. In this paper, we consider a aerial cooperative short packet communication system model in urban environments. The number of transmitted bits is maximized by jointly optimizing the UAV's altitude, transmit power and blocklength allocation with constraints on system reliability, latency and fairness. The successive convex approximation (SCA) method is adopted to solve the difficulty that the optimization problem is non-convex. Specifically, we firstly decompose the optimization problem into three subproblems to be solved separately, and then obtain the solution of the original problem by the proposed optimization algorithm. Finally, the performance of the proposed algorithm is verified by simulation results. In addition, we investigate the impact of different communication environments and the number of IoT devices on the system performance.

1. Introduction

The application of 6G communication network is expected to support the access of huge amount of smart devices and facilitate the further development of Internet of Things (IoT) [1–3]. With the help of connected smart devices, the IoT can efficiently perform complex missions and bring great convenience to human life [4,5]. Unmanned aerial vehicle (UAV) is considered as a key part of the air-ground network connectivity in 6G IoT [6–8]. In emergency communications, UAV can provide an effective communication solution because of its flexible deployment, low cost and high efficiency [9,10]. In addition, UAV can achieve line-of-sight (LoS) communication and provide efficient communication services for ground devices [11–13]. In recent years, the resource optimization problem in UAV communication has been studied to exploit the potential of UAV communication. In [14], the authors consider a UAV communication system that achieves green communication by optimizing the location and beamwidth of the UAV and the bandwidth of the ground terminal. In [15], the coexistence of device-to-device communication and UAV-assisted communication is considered, and the location and transmit power of the UAV are optimized to maximize the device-to-device communication rate. In [16], the secure transmission problem in UAV communication systems is considered and the authors maximize the minimum secure transmission rate by optimizing the UAV trajectory and transmit power. The problem of secure energy efficiency in UAV communication is studied in [17], where the UAV is equipped with reconfigurable intelligent surface and works as relay node. In [18], UAV is considered for emergency communication scenarios and then the communication rate of the system is improved by optimizing the UAV's altitude, transmit power and bandwidth allocation.

With the development of communication network technology, short packet transmission has become an important form of IoT application [19–21]. For example, mission-critical IoT scenarios and machine-to-machine communications employ short packets to transmit ultra-reliable and low-latency information [22,23]. The traditional communication theory is not applicable to short packet communication since it employs a finite blocklength to transmit information with non-negligible error probability. Over the years of research, significant results have been achieved in short packet communication theory. In [24], Polyanskiy et al. derived a mathematical expression for the maximum achievable rate for finite blocklength transmission in additive white Gaussian noise (AWGN) channels. On the basis of this study, [25,26] investigated the maximum achievable transmission rate under quasi-static multi-antenna fading channels and multi-antenna Rayleigh fading channels with finite blocklength, respectively. The design and optimization of short packet communication protocols for three specific communication scenarios are investigated in [27].

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Due to the unique advantages of UAV communication, UAV-enabled short packet communication has become a hot research topic. In [28], the resource optimization problem in UAV-assisted short packet communication is studied. In [29], the UAV location, transmission time and bandwidth allocation are optimized to improve the transmission rate of a two-way UAV relay short packet communication system. The reliability of UAV-enabled short packet relay communication systems adopting decode-and-forward mode is studied in [30]. Based on [30,31] investigates the optimization problem in amplified forwarding mode and adopts a three-dimensional channel model. In [32], the security of UAV short packet communication system is considered and the joint optimization of UAV trajectory and transmit power is performed to improve the secure transmission performance of the system. The power limitation problem of UAV-enabled IoT short packet communication networks is considered in [33], where the location of UAVs, the association among IoT devices and the transmit power are optimized to achieve short packet communication under limited power. A multi-hop UAV short packet communication system is considered in [34], and the reliability of the system is improved by jointly optimizing the block length allocation and the UAV position. [35] investigates the UAV-enabled uplink green short packet communication system.

So far, the research work on aerial cooperative short packet communication has mainly considered a pair of devices and multiple devices have not been sufficiently investigated, therefore we investigate aerial cooperative short packet communication with multiple devices, in addition, the fairness of information transmission among multiple devices is considered. In urban environments, air-to-ground (A2G) communication channel models have non-line-of-sight (NLoS) propagation caused by environmental factors (e.g., obstruction by high-rise buildings) in addition to LoS propagation. In this paper, we consider a model of aerial cooperative short packet communication system in which the probability-based LoS channel is adopted. Our goal is to maximize the number of information bits transmitted by the system in a transmission period, specifically, we achieve it by jointly optimizing the transmit power, blocklength allocation, and the altitude of the UAV that closely affect the amount of information transmitted. In the optimization problem of this paper, the reliability and low latency of the system are guaranteed, and the fairness of the relay communication is also considered. The main contributions of this paper are summarized as follows:

- According to the channel conditions in urban environments, probability-based LoS channels are adopted in the aerial cooperative short packet communication system. We formulate the optimization problem that the number of transmitted bits is maximized by jointly optimizing the transmit power, blocklength allocation and UAV's altitude with the guarantee of high reliability, low latency and fairness of the system.
- The optimization problem is difficult to be solved since the objective functions and some constraints are non-convex. We employ a successive convex approximation (SCA) method to solve the non-convexity of the optimization problem. Specifically, the optimization problem is firstly decomposed into three subproblems to be optimized separately, and then the proposed iterative optimization algorithm is used to solve the original problem.
- The simulation results show that the proposed optimization algorithm has low complexity. And the proposed algorithm has similar performance with the exhaustive method and outperforms other benchmark schemes. In addition, we also investigate the effects of different communication environments and the number of IoT devices on the performance of aerial cooperative short packet communication system.

The other parts of this paper are organized as follows. Section 2 presents the aerial cooperative short packet communication system model and mathematically formulates the optimization problem. The algorithm to solve the optimization problem is specified in Section 3. We present the simulation results and verify the performance of the proposed algorithm in Section 4. The paper is concluded in Section 5.

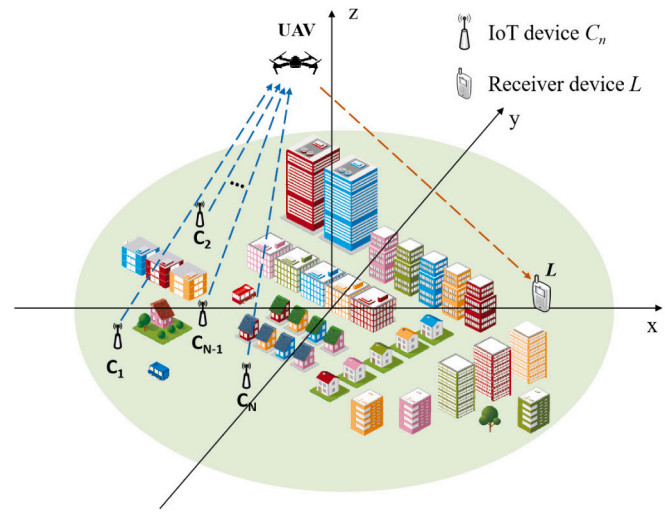

Fig. 1. Aerial cooperative short packet communication system.

2. System model and problem formulation

We consider a aerial cooperative short packet communication system that includes N IoT devices, a UAV and a receiver device L, which is shown in Fig. 1. There is no direct link between the IoT devices and the receiver device, therefore a UAV is used for relay communication. The transmission period is divided into two phases, the first phase is the transmission of short packets from N IoT devices to the UAV by using time division multiple access (TDMA) technology, and the second phase is the sending of the received short packets by the UAV to the receiver device L. At the same time, in order to guarantee the delay requirement of the system, the transmission process should be completed in t seconds, so the maximum blocklength is $M_{\max} = Bt$, where B is the system bandwidth. For the convenience of presentation, the nth IoT device is denoted as C_n, $n \in (1, 2, 3, \ldots, N)$, and the communication link from node i to node j is denoted as link-ij, $ij \in \{C_n U, UL\}$. Meanwhile, a three-dimensional Cartesian coordinate system is adopted to denote the position of the IoT device C_n as $\left(x_{C_n}, y_{C_n}, 0\right)$, the position of the UAV as (x_U, y_U, z), and the position of the receiver device L as $(x_L, y_L, 0)$. The key notation is summarized in Table 1.

An A2G communication path loss model for urban environments proposed in [36] is adopted, which considers mainly the LoS propagation and the NLoS propagation. The probability of occurrence of LoS communication is related to the elevation angle, and the elevation angle of link-ij is given as

$$\varphi_{ij} = \frac{180}{\pi} \arctan\left(\frac{z}{h_{ij}}\right), \tag{1}$$

where $h_{ij} = \sqrt{\left(x_i - x_j\right)^2 + \left(y_i - y_j\right)^2}$ is the horizontal distance of the link-ij. Then, The probability of LoS communication for link-ij is expressed as [36]

$$P_{ij}^{LoS} = \beta_0 \left(\varphi_{ij} - \varphi_0\right)^{\lambda_0}, \tag{2}$$

where β_0 and λ_0 are parameters determined by the environment, $0 < \beta_0 < 1$, $0 < \lambda_0 < 1$, φ_0 is the minimum elevation angle allowed for A2G communication. The probability of NLoS communication for link-ij is calculated as $P_{ij}^{NLoS} = 1 - P_{ij}^{LoS}$.

The channel gains of link-ij for LoS and NLoS are denoted as, respectively

$$\mu_{ij}^{LoS} = \alpha_0 d_{ij}^{-2} 10^{-0.1\kappa_{LoS}} \tag{3}$$

and

$$\mu_{ij}^{NLoS} = \alpha_0 d_{ij}^{-2} 10^{-0.1\kappa_{NLoS}}, \tag{4}$$

Table 1
Summary of key notation.

Notation	Meaning	Notation	Meaning
N	Numbers of IoT devices.	B	System bandwidth.
$M_{\max}$	Maximum blocklength.	C_n	The nth IoT device.
L	Receiver device.	U	Unmanned aerial vehicle.
z	Altitude of the UAV.	h_{ij}	Horizontal distance of the link-ij.
φ_{ij}	Elevation angle of link-ij.	β_0, λ_0	Parameters determined by the environment.
φ_0	Minimum elevation angle.	P_{ij}^{LoS}, P_{ij}^{NLoS}	LoS/NLoS probability for link-ij.
d_{ij}	Distance of link-ij.	α_0	Free-space channel gain for unit distance.
κ_{LoS}, κ_{NLoS}	Excessive path loss for LoS and NLoS.	μ_{ij}^{LoS}, μ_{ij}^{LoS}	Channel gains of link-ij for LoS and NLoS.
μ_{ij}	Average channel gain of link-ij.	R_{ij}	Maximum achievable rate of link-ij.
ε	Decoding error probability.	p_i	Transmit power of node i.
δ^2	Average power of the AWGN.	m_i	Blocklength allocated to node i.
γ_{ij}	SNR at node i for the signal sent by node j.	Γ_{ij}	Number of transmitted bits for link-ij.

where α_0 is the free-space channel gain for unit distance, $d_{ij} = \sqrt{h_{ij}^2 + z^2}$ is the distance of link-ij, κ_{LoS} and κ_{NLoS} are the excessive path loss (in dB) for LoS and NLoS, respectively. Therefore, the average channel gain of link-ij is expressed as follows

$$\mu_{ij} = P_{ij}^{LoS}\mu_{ij}^{LoS} + P_{ij}^{NLoS}\mu_{ij}^{NLoS} = d_{ij}^{-2}\left(P_{ij}^{LoS}K_1 + K_2\right), \tag{5}$$

where $K_1 = \alpha_0\left(10^{-0.1\kappa_{LoS}} - 10^{-0.1\kappa_{NLoS}}\right)$, $K_2 = \alpha_0 10^{-0.1\kappa_{NLoS}}$.

Different from long packet communication, the maximum achievable rate of short packet communication is lower than the Shannon capacity since the decoding error probability cannot be ignored. When the decoding error probability is ε, the maximum achievable rate of link-ij is given as [24]

$$R_{ij} = \log_2\left(1+\gamma_{ij}\right) - \sqrt{\frac{V_{ij}}{m_i}}\frac{Q^{-1}(\varepsilon)}{\ln 2}, \tag{6}$$

where $V_{ij} = 1 - \left(1+\gamma_{ij}\right)^{-2}$, $\gamma_{ij} = \frac{p_i\mu_{ij}}{\delta^2}$ is the signal-to-noise ratio at node i for the signal sent by node j, p_i is the transmit power of node i, δ^2 is the average power of the AWGN, m_i is the blocklength allocated to node i. $Q^{-1}(x)$ is the inverse function of $Q(x)$, where $Q(x) = \int_x^\infty \frac{1}{\sqrt{2\pi}}e^{-\frac{t^2}{2}}dt$. According to [24,37], the number of transmitted bits for link-ij is calculated as

$$\Gamma_{ij} = m_i R_{ij}. \tag{7}$$

In this paper, we aim to maximize the number of transmitted bits in a transmission period by optimizing the UAV's altitude z, the transmit power $\boldsymbol{p} = \left\{\left\{p_{C_n}, \forall n\right\} \cup p_U\right\}$ and the blocklength allocation $\boldsymbol{m} = \left\{\left\{m_{C_n}, \forall n\right\} \cup m_U\right\}$, in which the latency and reliability of the system and the fairness among multiple IoT devices are considered. The above optimization problem can be mathematically formulated as follows

$$\max_{z,\boldsymbol{p},\boldsymbol{m}} \quad \Gamma_{UL} \tag{8a}$$

$$\text{s.t.} \quad \Gamma_{UL} \le \sum_{n=1}^{N}\Gamma_{C_nU}, \tag{8b}$$

$$\Gamma_{C_1U} = \Gamma_{C_2U} = \cdots = \Gamma_{C_NU}, \tag{8c}$$

$$z \ge z_{\min}, \tag{8d}$$

$$\sum_{n=1}^{N} m_{C_n} + m_U \le M_{\max}, \tag{8e}$$

$$m_{C_n}, m_U \in \mathbb{N}, \quad \forall n \tag{8f}$$

$$0 \le p_{C_n} \le p_{\max}^{C}, \quad \forall n \tag{8g}$$

$$0 \le p_U \le p_{\max}^{U}, \tag{8h}$$

where constraint (8b) is the information causality constraint, i.e., the number of bits transmitted in the second phase is less than or equal to the number of bits transmitted in the first phase during a transmission period of the relay communication. (8c) is the constraint of fairness among IoT devices. $z_{\min}$ in (8d) is the minimum altitude of the UAV hovering. $M_{\max}$ in (8e) is the maximum blocklength available in a transmission period. (8f) means that m_{C_n} and m_U are non-negative integers. (8 g) and (8h) specify the range of transmit power for IoT devices and UAV.

Problem (8) formulated above is difficult to be solved for two reasons: (1) The objective function of problem (8) is complex since short packet communication and A2G communication path loss model for urban environments are considered. (2) The constraints (8b) and (8c) are nonconvex with respect to z, $\boldsymbol{p}$ and $\boldsymbol{m}$ which leads to a more challenging solution of problem (8).

3. Proposed algorithm

To facilitate the analysis, we introduce two auxiliary variables w and v to equivalently formulate problem (8) as follows

$$\max_{z,\mathbf{p},\mathbf{m},w,v} \quad w \tag{9a}$$

$$\text{s.t.} \quad (8\text{d}) - (8\text{h}), \tag{9b}$$

$$w \le \Gamma_{UL}, \tag{9c}$$

$$w \le Nv, \tag{9d}$$

$$v \le \Gamma_{C_nU}. \quad \forall n \tag{9e}$$

The objective function of problem (9) reaches the maximum when the equation in constraint (9c) holds and $\Gamma_{C_1U} = \Gamma_{C_2U} = \cdots = \Gamma_{C_NU} = v$. Therefore, problem (8) and problem (9) are equivalent, and we can obtain the optimal values by solving problem (9). Next, problem (9) will be divided into three subproblems to be solved.

3.1. UAV's altitude optimization

When the transmit power $\boldsymbol{p}$ and the blocklength allocation $\boldsymbol{m}$ are fixed, problem (9) is formulated as

$$\max_{z,w,v} \quad w \tag{10a}$$

$$\text{s.t.} \quad w \le \Gamma_{UL}(z), \tag{10b}$$

$$w \le Nv, \tag{10c}$$

$$v \le \Gamma_{C_nU}(z), \quad \forall n \tag{10d}$$

$$z \ge z_{\min}. \tag{10e}$$

The problem (10) is nonconvex since (10b) and (10d) are nonconvex constraints. To solve this difficulty, we adopt the SCA method to construct the approximate convex constraints for (10b) and (10d) in problem (10). We find that (10b) and (10d) have a similar form.

Therefore, we firstly deal with the nonconvexity of constraint (10b), and constraint (10d) will be solved in a same way.

Firstly, constraint (10b) is rewritten as follows

$$w \le m_U \log_2\left(1+\gamma_{UL}(z)\right) - \sqrt{m_U V_{UL}(z)}\frac{Q^{-1}(\varepsilon)}{\ln 2}. \tag{11}$$

By introducing auxiliary variables $\chi = \left\{\chi_{UL} \cup \left\{\chi_{C_nU}, \forall n\right\}\right\}$ and $\boldsymbol{\eta} = \left\{\eta_{UL} \cup \left\{\eta_{C_nU}, \forall n\right\}\right\}$,(11) can be replaced by the following three constraints:

$$\begin{aligned} w \le\ & m_U \log_2\left(1+\frac{p_U\chi_{UL}}{\delta^2}\right) \\ & - \sqrt{m_U}\frac{Q^{-1}(\varepsilon)}{\ln 2}\sqrt{1-(1+\frac{p_U\eta_{UL}}{\delta^2})^{-2}}, \end{aligned} \tag{12}$$

$$\chi_{UL} \le \mu_{UL}(z), \tag{13}$$

$$\eta_{UL} \ge \mu_{UL}(z). \tag{14}$$

Before analyzing the concavity of constraint (12), Theorem 1 is firstly provided.

Theorem 1. *We define*

$$f_1(\eta_{UL}) = \sqrt{m_U}\frac{Q^{-1}(\varepsilon)}{\ln 2}\sqrt{1-(1+\frac{p_U\eta_{UL}}{\delta^2})^{-2}} \tag{15}$$

and $f_1(\eta_{UL})$ is a concave function with respect to η_{UL}.

Proof. The first order derivative of $f_1(\eta_{UL})$ with respect to η_{UL} is calculated as follows

$$f_1'(\eta_{UL}) = \frac{p_U\sqrt{m_U}}{\delta^2}\frac{Q^{-1}(\varepsilon)}{\ln 2}A^{-\frac{1}{2}}D^{-3}, \tag{16}$$

where $A = 1-(1+\frac{p_U\eta_{UL}}{\delta^2})^{-2}$, $D = 1+\frac{p_U\eta_{UL}}{\delta^2}$. Then, the second order derivative of $f_1(\eta_{UL})$ is calculated as follows

$$f_1''(\eta_{UL}) = -\frac{p_U^2\sqrt{m_U}}{\delta^4}\frac{Q^{-1}(\varepsilon)}{\ln 2}(A^{-\frac{3}{2}}D^{-6}+3A^{-\frac{1}{2}}D^{-4}). \tag{17}$$

According to (17), it is easy to get $f_1''(\eta_{UL}) < 0$, hence $f_1(\eta_{UL})$ is a concave function with respect to η_{UL}. ■

Combined with Theorem 1, the right-hand side of (12) is easily proved to be a difference concave structure. Thus, the concave lower bound on the right-hand side of (12) can be obtained by the first-order Taylor expansion of $f_1(\eta_{UL})$. Then, we rewrite (12) as follows

$$\begin{aligned} w \le\ & m_U \log_2\left(1+\frac{p_U\chi_{UL}}{\delta^2}\right) - f_1(\eta_{UL}^k) \\ & - f_1'(\eta_{UL}^k)(\eta_{UL}-\eta_{UL}^k) = E_{UL}(\chi_{UL},\eta_{UL}), \end{aligned} \tag{18}$$

where η_{UL}^k is the point given at the kst iteration.

We introduce two auxiliary variables $\boldsymbol{\rho} = \left\{\rho_{UL} \cup \left\{\rho_{C_nU}, \forall n\right\}\right\}$ and $\boldsymbol{\tau} = \left\{\tau_{UL} \cup \left\{\tau_{C_nU}, \forall n\right\}\right\}$, and (13) is equivalently transformed into three constraints as follows

$$\tau_{UL} \ge \rho_{UL}\chi_{UL}, \tag{19}$$

$$\rho_{UL} \ge d_{UL}^2, \tag{20}$$

$$\tau_{UL} \le \beta_0(\varphi_{UL}-\varphi_0)^{\lambda_0}K_1 + K_2. \tag{21}$$

Among constraints (19)–(21), (20) is a convex constraint, while (19) and (21) are non-convex constraints. To deal with this problem, we firstly rewrite constraint (19) as follows

$$\tau_{UL}^{1/2} \ge (\rho_{UL}\chi_{UL})^{1/2}. \tag{22}$$

Since $(\rho_{UL}\chi_{UL})^{1/2}$ is a concave function, we can obtain the convex upper bound of $(\rho_{UL}\chi_{UL})^{1/2}$ by the first-order Taylor expansion. Therefore, the constraint (22) can be reformulated as

$$\begin{aligned} \tau_{UL}^{1/2} \ge\ & (\rho_{UL}^k\chi_{UL}^k)^{1/2} + \frac{1}{2}(\rho_{UL}^k)^{-1/2}(\chi_{UL}^k)^{1/2}(\rho_{UL}-\rho_{UL}^k) \\ & + \frac{1}{2}(\rho_{UL}^k)^{1/2}(\chi_{UL}^k)^{-1/2}(\chi_{UL}-\chi_{UL}^k) = F_{UL}(\rho_{UL},\chi_{UL}). \end{aligned} \tag{23}$$

In order to solve the nonconvexity problem of (21), constraint (21) is divided into four constraints by introducing auxiliary variables $\boldsymbol{s} = \left\{s_{UL} \cup \left\{s_{C_nU}, \forall n\right\}\right\}$ and ω as follows

$$\frac{1}{\omega} \le z, \tag{24}$$

$$s_{UL} \ge \varphi_0, \tag{25}$$

$$\tau_{UL} \le \beta_0(s_{UL}-\varphi_0)^{\lambda_0}K_1 + K_2, \tag{26}$$

$$s_{UL} \le \frac{180}{\pi}\arctan(\frac{1}{\omega h_{UL}}). \tag{27}$$

Among constraints (24)–(27), they are all convex constraints except constraint (27). Since $\arctan(1/\omega h_{UL})$ is a convex function with respect to ω, we can obtain a concave lower bound for $\arctan\left(1/\omega h_{UL}\right)$ by a first-order Taylor expansion. And thus the constraint (27) can be reformulated as

$$\begin{aligned} s_{UL} \le\ & \frac{180}{\pi}\arctan\left(\frac{1}{\omega^k h_{UL}}\right) \\ & - \frac{180}{\pi}\left(\frac{h_{UL}}{\left(\omega^k h_{UL}\right)^2+1}\right)\left(\omega-\omega^k\right) = G_{UL}(\omega). \end{aligned} \tag{28}$$

Constraint (14) perform the operation of taking the natural logarithm, and constraint (14) is rewritten as

$$\begin{aligned} \ln\eta_{UL} \ge\ & \ln\left(\beta_0\left(\frac{180}{\pi}\arctan\left(\frac{z}{h_{UL}}\right)-\varphi_0\right)^{\lambda_0}K_1+K_2\right) \\ & - \ln\left(h_{UL}^2+z^2\right). \end{aligned} \tag{29}$$

According to (24), we can transform the constraint (29) as follows

$$\begin{aligned} & \ln\eta_{UL} + \ln\left(h_{UL}^2+\frac{1}{\omega^2}\right) \\ & \ge \ln\left(\beta_0\left(\frac{180}{\pi}\arctan\left(\frac{z}{h_{UL}}\right)-\varphi_0\right)^{\lambda_0}K_1+K_2\right). \end{aligned} \tag{30}$$

Before analyzing the concavity of constraint (30), Theorem 2 is firstly provided.

Theorem 2. *Define*

$$f_2(\omega) = \ln\left(h_{UL}^2+\frac{1}{\omega^2}\right) \tag{31}$$

and

$$f_3(z) = \ln\left(\beta_0\left(\frac{180}{\pi}\arctan\left(\frac{z}{h_{UL}}\right)-\varphi_0\right)^{\lambda_0}K_1+K_2\right). \tag{32}$$

$f_2(\omega)$ is a convex function with respect to ω, and $f_3(z)$ is a concave function with respect to z.

Proof. Firstly, the first order derivative of $f_2(\omega)$ is calculated as follows

$$f_2'(\omega) = -\frac{2}{\omega^3 h_{UL}^2+\omega}. \tag{33}$$

Then the second order derivative of $f_2(\omega)$ is calculated as follows

$$f_2''(\omega) = \frac{6\omega^2 h_{UL}^2+2}{\left(\omega^3 h_{UL}^2+\omega\right)^2}. \tag{34}$$

According to (34), it is easy to obtain that $f_2''(\omega) > 0$, hence $f_2(\omega)$ is a convex function with respect to ω.

To prove that $f_3(z)$ is a concave function with respect to z, we define $g(z) = \frac{180}{\pi}\arctan\left(\frac{z}{h_{UL}}\right) - \varphi_0$ and the second order derivative of $g(z)$ is given by

$$g''(z) = -\frac{360 z h_{UL}}{\pi\left(z^2 + h_{UL}^2\right)^2}. \tag{35}$$

Based on the fact that $z > 0$, $h_{UL} > 0$, we conclude that $g''(z) < 0$. Therefore, $g(z)$ is a concave function with respect to z. Since $\ln y$ and $\beta_0 y^{\lambda_0} K_1 + K_2$ are concave and non-decreasing, we can easily prove that $f_3(z)$ is a concave function based on the compound rule. ■

According to Theorem 2, we can obtain the concave lower bound of $f_2(\omega)$ and the convex upper bound of $f_3(z)$ as follows

$$f_2(\omega) > f_2(\omega_k) + f_2'(\omega_k)(\omega - \omega_k) = H_{UL}(\omega), \tag{36}$$

$$f_3(z) < f_3(z_k) + f_3'(z_k)(z - z_k) = \Theta_{UL}(z), \tag{37}$$

where

$$f_3'(z) = \frac{\beta_0 \lambda_0 K_1 \left(g\left(z^k\right)\right)^{\lambda_0 - 1} \frac{180}{\pi} \frac{h_{UL}}{h_{UL}^2 + \left(z^k\right)^2}}{\beta_0 \left(g\left(z^k\right)\right)^{\lambda_0} K_1 + K_2}. \tag{38}$$

Combining (36) and (37), constraint (30) can be transformed to

$$\ln \eta_{UL} + H_{UL}(\omega) \geq \Theta_{UL}(z). \tag{39}$$

Based on the method of handling (10b), we can take the same steps to obtain the approximate convex constraint set of (10d). For consistency of presentation, we use a similar notation in solving the nonconvexity of the constraint (10d) as in dealing with the constraint (10b). For example, $E_{C_nU}\left(\chi_{C_nU}, \eta_{C_nU}\right)$ is used to represent an expression of the same form as $E_{UL}\left(\chi_{UL}, \eta_{UL}\right)$. Therefore, the approximate convex constraint set for (10d) is given as follows

$$v \leq E_{C_nU}\left(\chi_{C_nU}, \eta_{C_nU}\right), \quad \forall n \tag{40}$$

$$\rho_{C_nU} \geq d_{C_nU}^2, \quad \forall n \tag{41}$$

$$\tau_{C_nU}^{1/2} \geq F_{C_nU}\left(\rho_{C_nU}, \chi_{C_nU}\right), \quad \forall n \tag{42}$$

$$\tau_{C_nU} \leq \beta_0 \left(s_{C_nU} - \varphi_0\right)^{\lambda_0} K_1 + K_2, \quad \forall n \tag{43}$$

$$\varphi_0 \leq s_{C_nU} \leq G_{C_nU}(\omega), \quad \forall n \tag{44}$$

$$\ln \eta_{C_nU} + H_{C_nU}(\omega) \geq \Theta_{C_nU}(z), \quad \forall n \tag{45}$$

After the above analysis, problem (10) can be equivalently formulated by the use of the SCA method as

$$\max_{z,w,v,\omega,\boldsymbol{\rho},\boldsymbol{\tau},\boldsymbol{s},\boldsymbol{\chi},\boldsymbol{\eta}} \quad w \tag{46a}$$

$$\text{s.t.} \quad (8d), (9d)), (18), (20), \tag{46b}$$

$$(23) - (26), (28), (39) - (45). \tag{46c}$$

Obviously, problem (46) is a convex problem. Hence we can use CVX toolbox to solve problem (46) by the given initial parameters [38]. By iteratively optimizing problem (46), we can obtain the optimal UAV's altitude. The detailed optimization method is given in Algorithm 1.

Algorithm 1 : Optimization Algorithm for UAV's Altitude

1: Set $k = 0$ and initialize $\phi^k = \left\{z^k, \omega^k, \boldsymbol{\rho}^k, \boldsymbol{\chi}^k, \boldsymbol{\eta}^k\right\}$;
2: **Repeat**
3: Solve problem (46) by the given ϕ^k, and the solution to problem (46) is denoted as ϕ^{k+1};
4: $k = k + 1$;
5: **Until** convergence
6: $z^* = z^k$;
7: **Output** z^*.

3.2. Transmit power optimization

When the UAV's altitude z and blocklength allocation $\boldsymbol{m}$ are fixed, problem (9) is formulated as

$$\max_{\boldsymbol{p},w,v} \quad w \tag{47a}$$

$$\text{s.t.} \quad w \leq \Gamma_{UL}(p_U), \tag{47b}$$

$$w \leq N v, \tag{47c}$$

$$v \leq \Gamma_{C_nU}\left(p_{C_n}\right), \quad \forall n \tag{47d}$$

$$0 \leq p_{C_n} \leq p_{\max}^C, \quad \forall n \tag{47e}$$

$$0 \leq p_U \leq p_{\max}^U. \tag{47f}$$

Since (47b) and (47d) are non-convex constraints, problem (47) is a non-convex problem. We define $\Gamma_{UL}(p_U) = h_1(p_U) - h_2(p_U)$ and the expressions for $h_1(p_U)$ and $h_2(p_U)$ are given as follows

$$h_1(p_U) = m_U \log_2\left(1 + \gamma_{UL}(p_U)\right), \tag{48}$$

$$h_2(p_U) = \sqrt{m_U V_{UL}(p_U)} \frac{Q^{-1}(\varepsilon)}{\ln 2}, \tag{49}$$

$h_1(p_U)$ and $h_2(p_U)$ are easily proved to be concave functions with respect to p_U respectively. Therefore, $\Gamma_{UL}(p_U)$ is a function with a concave difference structure, and the concave lower bound of $\Gamma_{UL}(p_U)$ is given as follows

$$\begin{aligned}\Gamma_{UL}(p_U) \geq\ & h_1\left(p_U^k\right) - h_2\left(p_U^k\right) \\ & - h_2'\left(p_U^k\right)\left(p_U - p_U^k\right) = \Gamma_{UL}^{lb}(p_U),\end{aligned} \tag{50}$$

where

$$h_2'\left(p_U^k\right) = \frac{\mu_{UL}}{\delta^2}\sqrt{\frac{m_U}{V_{UL}\left(p_U^k\right)}} \frac{Q^{-1}(\varepsilon)}{\ln 2}\left(1 + \gamma_{UL}\left(p_U^k\right)\right)^{-3}.$$

Similar to $\Gamma_{UL}(p_U)$, the concave lower bound of $\Gamma_{C_nU}\left(p_{C_n}\right)$ in constraint (9e) is given as follows

$$\begin{aligned}\Gamma_{C_nU}\left(p_{C_n}\right) \geq\ & h_1\left(p_{C_n}^k\right) - h_2\left(p_{C_n}^k\right) \\ & - h_2'\left(p_{C_n}^k\right)\left(p_{C_n} - p_{C_n}^k\right) = \Gamma_{C_nU}^{lb}\left(p_{C_n}\right),\end{aligned} \tag{51}$$

where

$$h_2'\left(p_{C_n}^k\right) = \frac{\mu_{C_nU}}{\delta^2}\sqrt{\frac{m_{C_n}}{V_{C_nU}\left(p_{C_n}^k\right)}} \frac{Q^{-1}(\varepsilon)}{\ln 2}\left(1 + \gamma_{C_nU}\left(p_{C_n}^k\right)\right)^{-3}.$$

According to (50) and (51), Problem (47) can be equivalently transformed into

$$\max_{\boldsymbol{p},w,v} \quad w \tag{52a}$$

$$\text{s.t.} \quad (47c), (47e), (47f), \tag{52b}$$

$$w \leq \Gamma_{UL}^{lb}(p_U), \tag{52c}$$

$$v \leq \Gamma_{C_nU}^{lb}\left(p_{C_n}\right), \quad \forall n \tag{52d}$$

Obviously, problem (52) is a convex problem. Hence we can use CVX toolbox to solve problem (52) by the given initial parameters. By iteratively optimizing the problem (52), we can obtain the optimal transmit power. The detailed optimization method is given in Algorithm 2.

Algorithm 2 : Optimization Algorithm for Transmit Power

Set $k = 0$ and initialize $\boldsymbol{p}^k$;
Repeat
Solve problem (52) by the given $\boldsymbol{p}^k$, and the solution to problem (52) is denoted as $\boldsymbol{p}^{k+1}$;
$k = k + 1$;
Until convergence
$\boldsymbol{p}^* = \boldsymbol{p}^k$;
Output $\boldsymbol{p}^*$.

3.3. Blocklength allocation optimization

When the UAV's altitude z and the transmit power $\boldsymbol{p}$ are fixed, problem (9) is formulated as

$$\max_{\boldsymbol{m},w,v} \quad w \tag{53a}$$
$$\text{s.t.} \quad w \le \Gamma_{UL}\left(m_U\right), \tag{53b}$$
$$w \le Nv, \tag{53c}$$
$$v \le \Gamma_{C_nU}\left(m_{C_n}\right), \quad \forall n \tag{53d}$$
$$\sum_{n=1}^{N} m_{C_n} + m_U \le M_{\max}, \tag{53e}$$
$$m_{C_n}, m_U \in \mathbb{N}. \quad \forall n \tag{53f}$$

Since variable $\boldsymbol{m}$ is a non-negative set of integers, problem (53) is a non-continuous variable optimization. To solve this difficulty, we firstly relax the constraint (53f) to a continuous variable constraint, so that problem (53) can be transformed as follows

$$\max_{\boldsymbol{m},w,v} \quad w \tag{54a}$$
$$\text{s.t.} \quad (53\text{b}) - (53\text{e}), \tag{54b}$$
$$m_{C_n} \ge 0, m_U \ge 0. \quad \forall n \tag{54c}$$

The constraints (53b) and (53d) in problem (54) are non-convex constraints. The right-hand sides of (53b) and (53d), i.e., $\Gamma_{UL}\left(m_U\right)$ and $\Gamma_{C_nU}\left(m_{C_n}\right)$, can easily be proved to be convex functions with respect to m_U and m_{C_n}, respectively. Thus, the first-order Taylor expansion is employed to obtain concave lower bounds for $\Gamma_{UL}\left(m_U\right)$ and $\Gamma_{C_nU}\left(m_{C_n}\right)$, respectively, as follows

$$\Gamma_{UL}\left(m_U\right) \ge \Gamma_{UL}\left(m_U^k\right) + \Gamma'_{UL}\left(m_U^k\right)\left(m_U - m_U^k\right), \tag{55}$$

$$\Gamma_{C_nU}\left(m_{C_n}\right) \ge \Gamma_{C_nU}\left(m_{C_n}^k\right) + \Gamma'_{C_nU}\left(m_{C_n}^k\right)\left(m_{C_n} - m_{C_n}^k\right), \tag{56}$$

where

$$\Gamma'_{UL}\left(m_U^k\right) = \log_2\left(1+\gamma_{UL}\right) - \frac{1}{2}\sqrt{\frac{V_{UL}}{m_U^k}}\frac{Q^{-1}(\varepsilon)}{\ln 2}, \tag{57}$$

$$\Gamma'_{C_nU}\left(m_{C_n}^k\right) = \log_2\left(1+\gamma_{C_nU}\right) - \frac{1}{2}\sqrt{\frac{V_{C_nU}}{m_{C_n}^k}}\frac{Q^{-1}(\varepsilon)}{\ln 2}. \tag{58}$$

According to (55) and (56), problem (54) can be equivalently transformed into

$$\max_{\boldsymbol{m},w,v} \quad w \tag{59a}$$
$$\text{s.t.} \quad (53\text{c}), (53\text{e}), (54\text{c}), \tag{59b}$$

Algorithm 3 : Optimization Algorithm for Blocklength Allocation

Set $k = 0$ and initialize $\boldsymbol{m}^k = \left\{\left\{m_{C_n}^k, \forall n\right\} \cup m_U^k\right\}$;
Repeat
Solve problem (59) by the given $\boldsymbol{m}^k$, and the solution to problem (59) is denoted as $\boldsymbol{m}^{k+1}$;
$k = k + 1$;
Until convergence
$\boldsymbol{m}^* = \boldsymbol{m}^k$;
$b = M_{\max} - \left(\sum_{n=1}^{N}\left\lfloor m_{C_n}^*\right\rfloor + \left\lfloor m_U^*\right\rfloor\right)$;
for $n = 1 : b$
$m_{C_n}^* = m_{C_n}^* + 1$;
end
Output $\boldsymbol{m}^*$.

$$w \le \Gamma_{UL}\left(m_U^k\right) + \Gamma'_{UL}\left(m_U^k\right)\left(m_U - m_U^k\right), \tag{59c}$$
$$v \le \Gamma_{C_nU}\left(m_{C_n}^k\right) + \Gamma'_{C_nU}\left(m_{C_n}^k\right)\left(m_{C_n} - m_{C_n}^k\right). \quad \forall n \tag{59d}$$

Problem (59) is a convex problem. Hence we can use CVX toolbox to solve problem (59) by the given initial parameters. When the blocklength $\boldsymbol{m}$ is considered as a continuous variable, we can obtain the optimal blocklength allocation $\boldsymbol{m}^*$ by solving problem (59). To satisfy the requirement that $\boldsymbol{m}^*$ is a non-negative integer, we firstly use the floor function to perform an integer operation on $\boldsymbol{m}^*$, and then allocate the unused blocklength according to the channel gain under the constraint of the maximum blocklength $M_{\max}$. For ease of presentation, IoT devices are ranked according to the average channel gain, i.e. $\mu_{C_1U} \ge \mu_{C_2U} \ge \cdots \ge \mu_{C_NU}$. The detailed optimization procedure is given in Algorithm 3.

3.4. Joint optimization algorithm and analysis

Combining the optimization algorithms of the three subproblems, we propose Algorithm 4 to obtain the solution of problem (8). In the $(l+1)$th iteration of Algorithm 4, the change in the value of the objective function for problem (8) is given as follows

$$\begin{aligned}\Gamma_{UL}\left(z^l, \boldsymbol{p}^l, \boldsymbol{m}^l\right) &\le \Gamma_{UL}\left(z^{l+1}, \boldsymbol{p}^l, \boldsymbol{m}^l\right) \le \Gamma_{UL}\left(z^{l+1}, \boldsymbol{p}^{l+1}, \boldsymbol{m}^l\right) \\ &\le \Gamma_{UL}\left(z^{l+1}, \boldsymbol{p}^{l+1}, \boldsymbol{m}^{l+1}\right).\end{aligned} \tag{60}$$

Since Γ_{UL} is nondecreasing in iterative optimization and the optimal solution of problem (8) is the upper bound of the iterative optimization result, the convergence of Algorithm 4 is guaranteed. Before analyzing the complexity of Algorithm 4, we analyze the complexity of the optimization algorithm for the three subproblems. The complexity of Algorithm 1 is $T_1 = \mathcal{O}\left(k_1(5N+9)^{3.5}\right)$, where $5N+9$ is the number of variables of problem (46) and k_1 is the number of iterations of Algorithm 1. Similar to Algorithm 1, the complexity of Algorithm 2 and Algorithm 3 are $T_2 = \mathcal{O}\left(k_2(N+3)^{3.5}\right)$ and $T_3 = \mathcal{O}\left(k_3(N+3)^{3.5}\right)$, respectively. Therefore, the complexity of Algorithm 4 is $T = \mathcal{O}\left(l\left(T_1+T_2+T_3\right)\right)$.

4. Simulation results

In this section, we verify the performance of the proposed optimization algorithm through extensive simulations. For the aerial cooperative short packet communication system model considered in this paper, we assume that there are $N = 3$ IoT devices (C_1, C_2, C_3) which are located at coordinates $(-500, 0, 0)$, $(-700, 30, 0)$ and $(-800, -30, 0)$. The receiver device L is located at coordinate $(800, 0, 0)$. The UAV is located between the IoT device and the receiver device L with $x_U = 0$ and $y_U = 0$. The minimum altitude of the UAV hovering is $z_{\min} = 230$ m. The system bandwidth is $B = 800$ KHz. The maximum transmission delay is $t = 1$ ms. Therefore, $M_{\max} = Bt = 800$. We consider four different communication environments (Suburban, Urban, Dense Urban,

Algorithm 4 : Optimized Algorithms for Joint Altitude and Resource Allocation

Set $l = 0$ and initialize $z^l, \boldsymbol{p}^l, \boldsymbol{m}^l$;
Repeat
For the given $\boldsymbol{p}^l$ and $\boldsymbol{m}^l$, the optimal altitude of the UAV z^{l+1} is obtained by using algorithm 1;
For the given z^{l+1} and $\boldsymbol{m}^l$, the optimal transmit power $\boldsymbol{p}^{l+1}$ is obtained by using algorithm 2;
For the given z^{l+1} and $\boldsymbol{p}^{l+1}$, the optimal blocklength allocation $\boldsymbol{m}^{l+1}$ is obtained by using algorithm 3;
$l = l + 1$;
Until convergence
$z^* = z^l$, $\boldsymbol{p}^* = \boldsymbol{p}^l$, $\boldsymbol{m}^* = \boldsymbol{m}^l$;
Output z^*, $\boldsymbol{p}^*$, $\boldsymbol{m}^*$.

Table 2
Parameters in different communication environments.

	κ_{LoS}	κ_{NLoS}	β_0	λ_0
Suburban	0.1	21	0.76	0.06
Urban	1.0	20	0.60	0.11
Dense Urban	1.6	23	0.36	0.21
Highrise Urban	2.3	34	0.05	0.61

Table 3
Simulation parameters.

Parameter	Value
Maximum transmit power of the IoT device, $p^C_{\max}$	1 W
Maximum transmit power of the UAV, $p^U_{\max}$	5 W
System decoding error probability, ε	10^{-6}
Minimum elevation angle, φ_0	15°
Free-space unit channel gain, α_0	−60 dB
Average power of the AWGN, δ^2	−160 dBm

Table 4
Algorithm execution time.

Algorithm	Executing time
Exhaustive	2617.47 s
Proposed	43.26 s
ADMM-Based	69.14 s
JOPB	12.38 s
JOPB+F	8.54 s
JOAB+F	31.89 s
JOAP+F	26.12 s

Highrise Urban) for the channel in the simulation. The parameters for the different communication environments are given in Table 2 [36]. If not specified, parameters are selected for the urban environment. Meanwhile, the other parameters to be used in the simulation are given in Table 3.

We propose six schemes to compare with the scheme proposed in this paper. The six schemes are as follows: (1) Exhaustive scheme: An exhaustive search method is used to solve the optimization problem. (2) ADMM-Based scheme: The alternating direction method of multipliers is adopted to jointly optimize the UAV's altitude, transmit power and blocklength allocation [39]. (3) JOPB scheme: Joint optimization of transmit power and blocklength allocation without considering fairness among IoT devices [28]. (4) JOPB+F scheme: Joint optimization of transmit power and blocklength allocation with consideration of fairness among IoT devices. (5) JOAB+F scheme: Joint optimization of UAV's altitude and blocklength allocation with consideration of fairness among IoT devices. (6) JOAP+F scheme: Joint optimization of UAV's altitude and transmit power with consideration of fairness among IoT devices. The execution times of all the algorithms are given in Table 4.

Fig. 2 shows the convergence procedures of Algorithm 4 for different decoding error probabilities. We can see that as the decoding error probability required by the system decreases, the number of

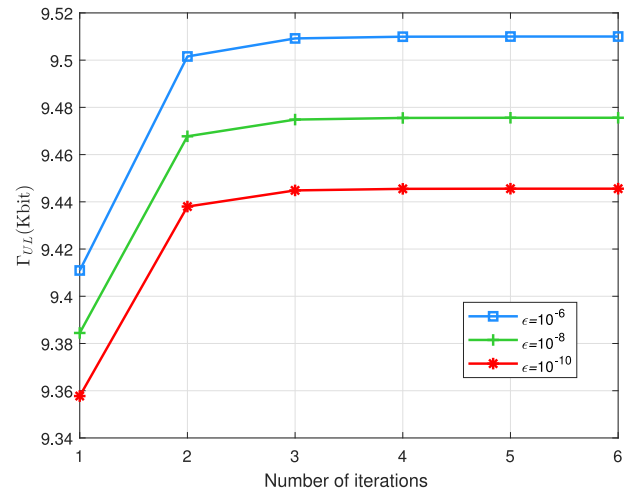

Fig. 2. The convergence procedures of Algorithm 4 for different values of ε.

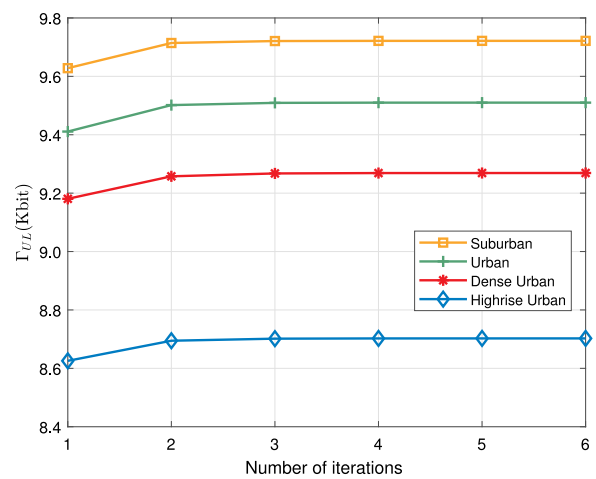

Fig. 3. The convergence procedure of Algorithm 4 for different communication environments.

bits transmitted decreases as well. Therefore, in order to ensure the reliability and effectiveness of aerial cooperative short packet communication system, it is necessary to choose an appropriate value of decoding error probability. Fig. 3 depicts the convergence procedure of Algorithm 4 for different communication environments. Figs. 2 and 3 show that Algorithm 4 can achieve stable and fast convergence. And the shown convergence procedure usually reaches convergence after four iterations, thus Algorithm 4 has low complexity.

Fig. 4 analyzes the blocklength allocation versus the maximum blocklength $M_{\max}$. It can be seen that the maximum blocklength $M_{\max}$ is fully utilized by the system and the blocklength allocated to each node increases linearly with the maximum blocklength $M_{\max}$. Specifically, the UAV is required to transmit the information received from multiple IoT devices to L, so the UAV is allocated nearly half of $M_{\max}$. Fig. 4 shows that the C_3 with lower average channel gain is allocated more blocklengths than C_2 and C_1. The reason is that the fairness of information transmission among the IoT devices should be guaranteed.

Fig. 5 plots the optimal altitude of the UAV for different communication environments (Suburban, Urban, Dense Urban, Highrise Urban). We have marked the locations of the IoT devices and the receiver device L with green dots and orange circles, respectively. The minimum altitude position of the UAV hovering is indicated by a gray cross. As shown in Fig. 5, we employ the proposed algorithm to obtain the

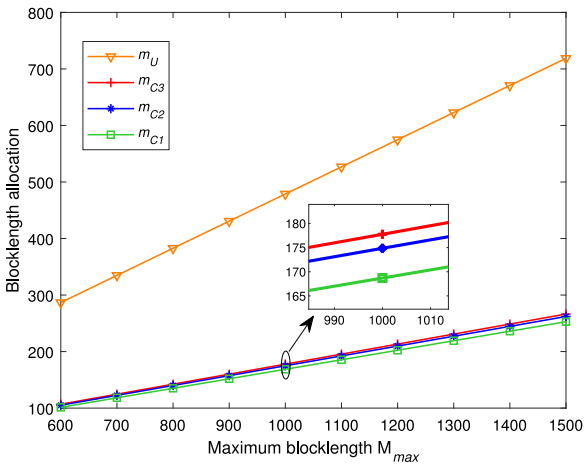

Fig. 4. The blocklength allocation versus the maximum blocklength $M_{\max}$.

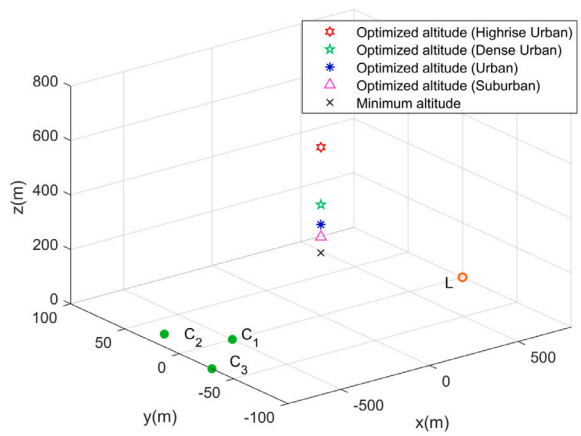

Fig. 5. The optimal altitude of the UAV for different communication environments.

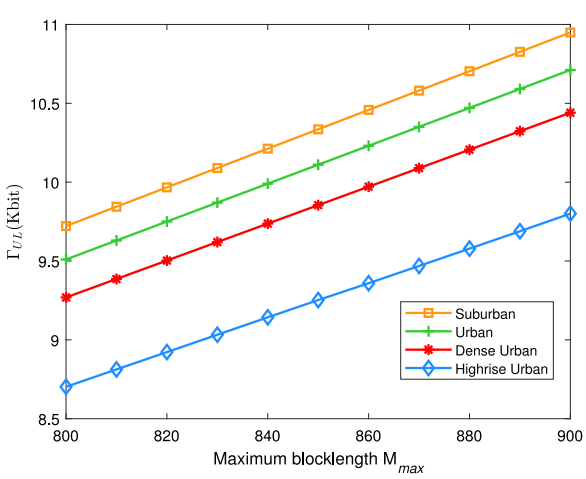

Fig. 6. Γ_{UL} versus the maximum blocklength $M_{\max}$.

optimal altitude of UAV hovering in different communication environments. As the buildings in the urban area become denser and taller, the UAV needs to hover at a higher position to improve the probability of LoS communication.

Fig. 6 shows the number of bits transmitted Γ_{UL} versus the maximum blocklength $M_{\max}$ for different communication environments. The number of transmitted bits Γ_{UL} increases linearly with the maximum

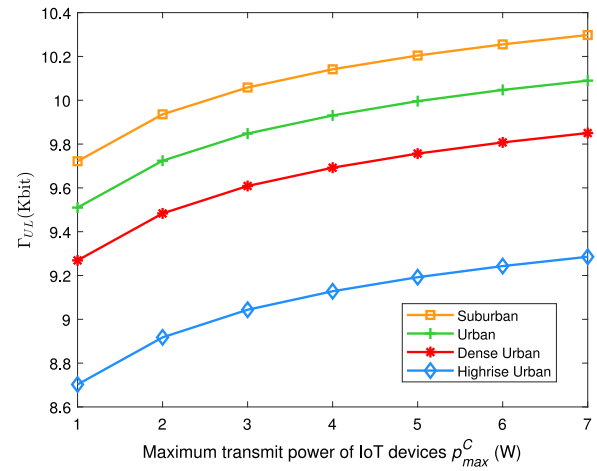

Fig. 7. Γ_{UL} versus the maximum transmit power $p^C_{\max}$ of the IoT devices.

blocklength $M_{\max}$. When the system bandwidth is given, the value of $M_{\max}$ is determined by the maximum delay of the system, which indicates there is a trade-off between the number of transmitted bits and the transmission delay of the system. In addition, since the suburban environment has better channel quality, more amount of information can be transmitted with the same value of $M_{\max}$ for the suburban environment.

Fig. 7 depicts the number of bits of information transmitted Γ_{UL} versus the maximum transmit power $p^C_{\max}$ of the IoT devices. The number of information bits transmitted by the system can be improved by increasing the value of $p^C_{\max}$. Also, we can find that the growth rate of Γ_{UL} decreases slowly with the value of $p^C_{\max}$. In addition, since the suburban environment has better channel quality, more information bits can be transmitted with the same value of $p^C_{\max}$ for the suburban environment.

Fig. 8 analyzes the relationship between the number of bits transmitted by each link and the number of IoT devices N in different communication environments. In the simulation of Fig. 8, the IoT devices are always uniformly distributed within a certain range. As more IoT devices share communication resources, fewer resources are allocated to each IoT device, which leads to a decrease in the number of bits transmitted from each IoT device to the UAV. Γ_{UL} decreases with the value of N. This is because the increase of N leads to a decrease in the blocklength allocated to each IoT device, and according to (6), the short packet communication rate of each link decreases in the first phase of relay communication, which leads to a decrease of Γ_{UL} in the second phase.

Fig. 9 presents the number of transmitted bits on each link for different schemes. Both the proposed scheme and JOPB+F scheme consider the fairness constraints of the uplink communication, and the proposed scheme can obtain better communication performance than JOPB+F scheme, which indicates the importance of the UAV's altitude optimization. As shown in Fig. 9, the information bits transmitted by the system in the JOPB scheme are mainly transmitted by link C_1U. This is because link C_1U has better channel quality, and is allocated more communication resources to maximize the transmission of information bits. This leads to the fact that the efficient transmission of information on each link cannot be guaranteed. Based on the above analysis, the proposed scheme has better system performance and can also guarantee the fairness of the transmission of each link.

Fig. 10 compares the performance of the six schemes for the various $p^C_{\max}$. The proposed scheme achieves almost the same performance with the exhaustive method for different $p^C_{\max}$, and the performance of the proposed scheme is better than other schemes. In addition, we can find that the performance of JOPB scheme is better than JOPB+F scheme.

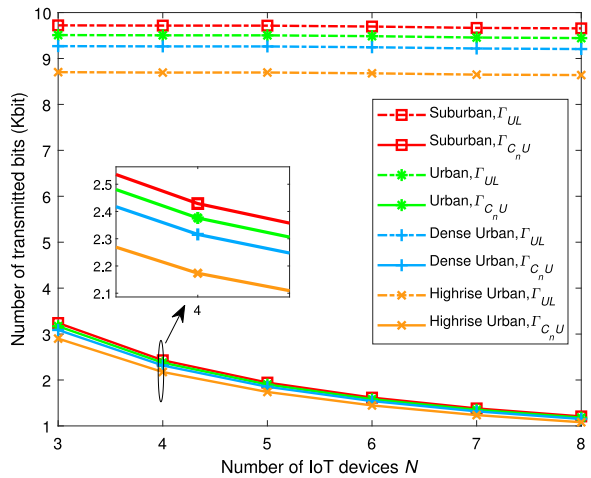

Fig. 8. Number of transmitted bits versus the number of IoT devices N.

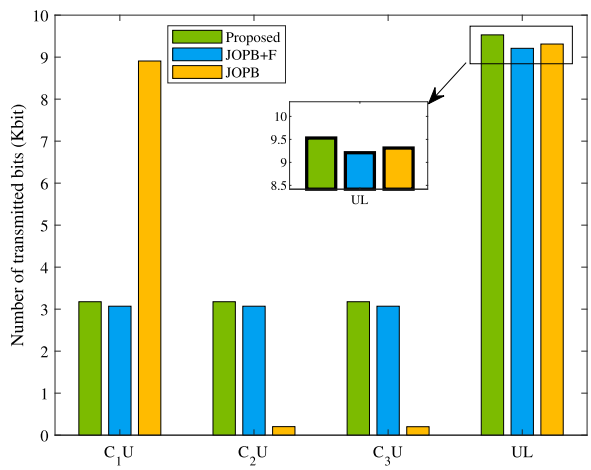

Fig. 9. The number of transmitted bits on each link for different schemes.

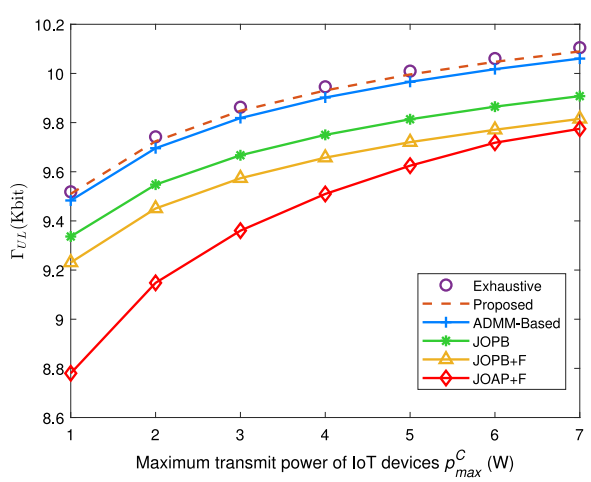

Fig. 10. The performance of the six schemes for the various values of $p^C_{\max}$.

This is because the JOPB scheme does not consider the fairness among the IoT devices, and the IoT devices with better channel quality are allocated more resources in order to transmit more information bits. Fig. 11 shows the number of bits transmitted Γ_{UL} versus the maximum blocklength $M_{\max}$ for different schemes. The proposed scheme achieves

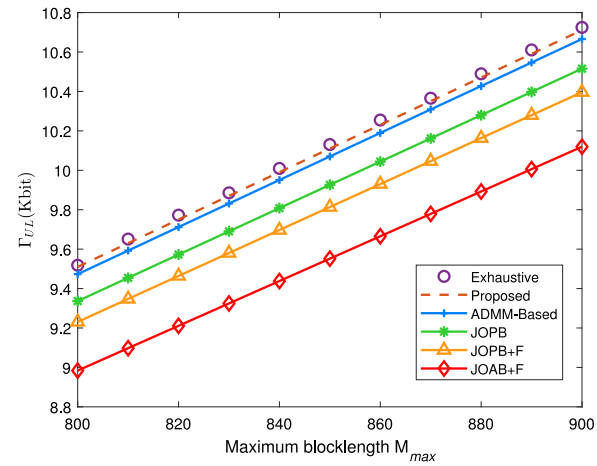

Fig. 11. The performance of the six schemes for the various values of $M_{\max}$.

almost the same performance with the exhaustive method for different $M_{\max}$ and the performance of the proposed scheme is better than other schemes.

5. Conclusions

In this paper, we study the problem of maximizing the number of information bits transmitted in a transmission period for the aerial cooperative short packet communication system. To facilitate the analysis, the optimization problem is formulated mathematically, then we propose a joint optimization algorithm to solve the optimization problem. Finally, the simulation results verify the performance of the proposed algorithm. In addition, we also investigate the effects of different communication environments and the number of IoT devices on the system performance. We find that the communication system can achieve better performance in the Suburban compared to other considered environments. As the buildings in the urban area become denser and taller, the UAV needs to hover at a higher position to improve the probability of LoS communication. And the increase in IoT devices will lead to a decrease in the performance of the communication system.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

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